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📐 MATHEMATICS PART 2 (GEOMETRY) | CLASS 10 | MAHARASHTRA SSC BOARD

Chapter 4: Geometric Constructions — Construction of Similar Triangle

TOPIC QUESTION SET · 13 QUESTIONS · TAUGHT 10 OCT 2026

🧭 TAUGHT TODAY · TEXTBOOK PAGES 91–95

- Similar triangles: corresponding sides proportional, corresponding angles equal
- Finding sides of the new triangle from the given ratio (Ex. 1)
- Construction when vertices are distinct (Ex. 1)
- Common vertex — smaller triangle, ratio 5 : 3 (Ex. 2)
- Common vertex — bigger triangle, ratio 5 : 7 (Ex. 3)
- Dividing a segment into equal parts; line parallel through a point

📊 WHERE THE QUESTIONS COME FROM

Source	Qs
📘 SSC Textbook — Problem Set 4	3
🗂️ Answer Bank — Ch 4	10
📗 NCERT Textbook folder	0

⚠️ NCERT: your folder holds the current Class 10 NCERT book (14 chapters). It has no constructions chapter, so nothing matched.
🚫 Left out: Practice Set 4.1 (Q1–4) and its copy in the Answer Bank (Q.3 (B) Q15).

A 📘 From SSC Textbook — Problem Set 4

3 Q

Balbharati Math-2, page 99 · Practice Set 4.1 questions left out as asked

📘 Textbook · Problem Set 4 · Q.1 (3) · Page 99

Q1 **If ΔABC ~ ΔPQR and $\frac{AB}{PQ} = \frac{7}{5}$, then .....**

(A) Δ ABC is bigger. **(B)** Δ PQR is bigger.
(C) Both triangles will be equal. **(D)** Can not be decided.

📘 Textbook · Problem Set 4 · Q.7 · Page 99

Q2 **Δ ABC ~ ΔLBN. In Δ ABC, AB = 5.1cm, ∠B = 40°, BC=4.8 cm, $\frac{AC}{LN} = \frac{4}{7}$. Construct Δ ABC and ΔLBN.**

📘 Textbook · Problem Set 4 · Q.8 · Page 99

Q3 **Construct Δ PYQ such that, PY = 6.3 cm, YQ = 7.2 cm, PQ = 5.8 cm. If $\frac{YZ}{YQ} = \frac{6}{5}$, then construct Δ XYZ similar to Δ PYQ.**

B 🎯 Answer Bank — Multiple Choice

4 Q

Answer Bank Ch 4 · Q.1 (A)

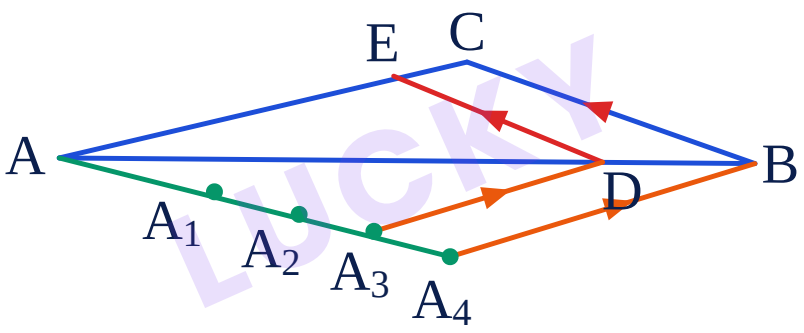
🗂️ Answer Bank · Q.1 (A) · Q3

Q4 **ΔLMN ~ ΔHIJ and $\frac{LM}{HI} = \frac{2}{3}$ then**

a ΔLMN is a smaller triangle. **b** ΔHIJ is a smaller triangle.
c Both triangles are congruent. **d** Can't say.

🗂️ Answer Bank · Q.1 (A) · Q5

Q5 **In the figure ΔABC ~ ΔADE then the ratio of their corresponding sides is:**



Figure

a $\frac{3}{1}$ **b** $\frac{1}{3}$

c $\frac{3}{4}$ **d** $\frac{4}{3}$

🗂️ Answer Bank · Q.1 (A) · Q7

Q6 **ΔPQR ~ ΔABC, $\frac{PR}{AC} = \frac{5}{7}$, then**

a ΔABC is greater. **b** ΔPQR is greater.

c Both triangles are congruent. **d** Can't say.

🗂️ Answer Bank · Q.1 (A) · Q8

Q7 **ΔABC ~ ΔAQR. $\frac{AB}{AQ} = \frac{7}{5}$, then which of the following option is true?**

a A-Q-B **b** A-B-Q

c A-C-B **d** A-R-B

C 📐 Answer Bank — Construct Similar Triangles

6 Q

Answer Bank Ch 4 · Q.3 (B)

🗂️ Answer Bank · Q.3 (B) · Q1

Q8 **ΔABC ~ ΔPBQ. In ΔABC, AB = 3 cm, ∠B = 90°, BC = 4 cm. Ratio of the corresponding sides of two triangles is 7 : 4. Then construct ΔABC and ΔPBQ.**

🗂️ Answer Bank · Q.3 (B) · Q2 *(repeated as Q13)*

Q9 **ΔRHP ~ ΔNED, In ΔNED, NE = 7 cm, ∠D = 30°, ∠N = 20°, $\frac{HP}{ED} = \frac{4}{5}$, then construct ΔRHP and ΔNED.**

🗂️ Answer Bank · Q.3 (B) · Q3

Q10 **ΔPQR ~ ΔABC. In ΔPQR, PQ = 3.6 cm, QR = 4 cm, PR = 4.2 cm. Ratio of the corresponding sides of triangle is 3 : 4, then construct ΔPQR and ΔABC.**

🗂️ Answer Bank · Q.3 (B) · Q4

Q11 Construct an equilateral ΔABC with side 5 cm. ΔABC ~ ΔLMN, ratio the corresponding sides of triangle is 6 : 7, then construct ΔLMN and ΔABC.

🗂️ Answer Bank · Q.3 (B) · Q12

Q12 ΔAMT ~ ΔAHE. In ΔAMT, AM = 6.3 cm, ∠MAT = 120°, AT = 4.9 cm, $\frac{AM}{HA} = \frac{7}{5}$, then construct ΔAMT and ΔAHE.

🗂️ Answer Bank · Q.3 (B) · Q14

Q13 ΔABC ~ ΔPBR, BC = 8 cm, AC = 10 cm, ∠B = 90°, $\frac{BC}{BR} = \frac{5}{4}$ then construct ΔABC and ΔPBR.

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